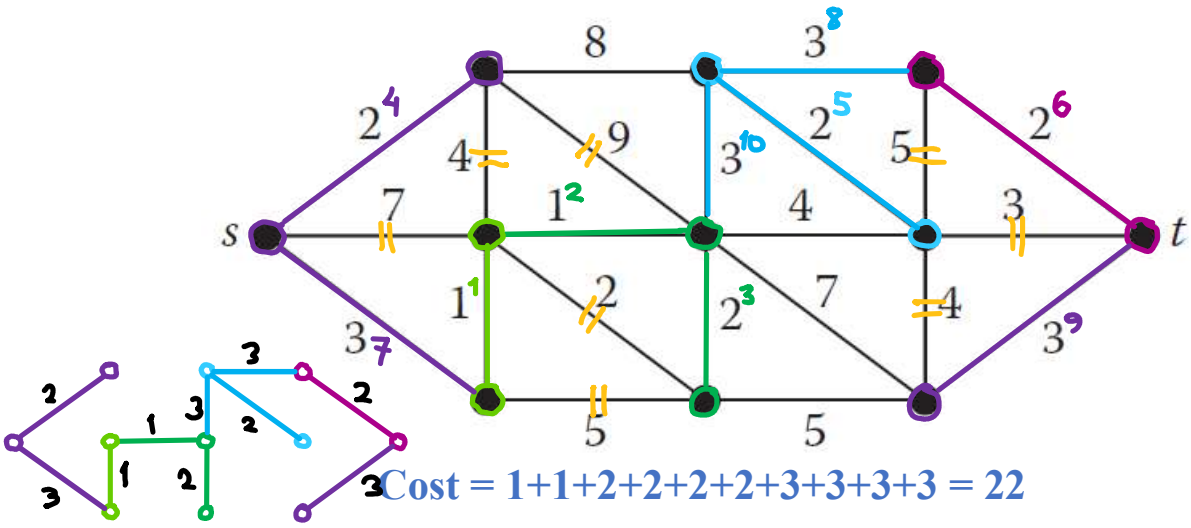
CSE 5004-Graph Theory and Algorithms

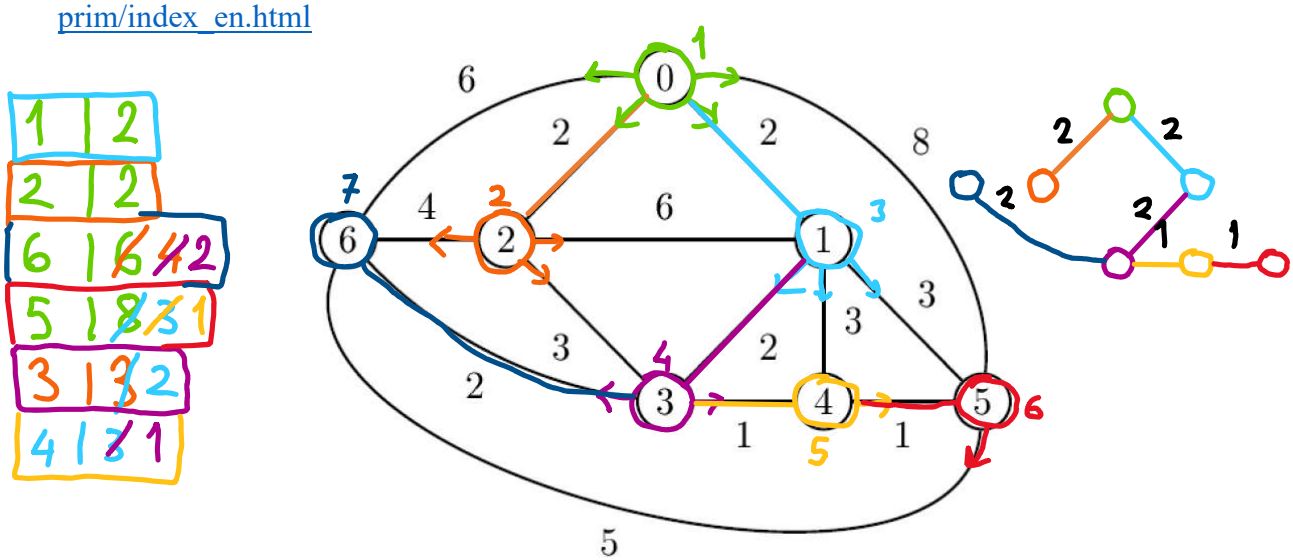
Exercise – 4

1. For the graph given below use Kruskal's algorithm to find a minimum spanning tree.
Hint: You can use the tool at https://algorithms.discrete.ma.tum.de/graph-algorithms/mst-kruskal/index_en.html



Cost = 1+1+2+2+2+2+3+3+3+3 = 22

2. For the following graph use Prim's algorithm to find a minimum spanning tree.
Hint: You can use the tool at https://algorithms.discrete.ma.tum.de/graph-algorithms/mst-prim/index_en.html



Cost = 2+2+2+1+1+2 = 10

3. HA30 dataset describes the airline distances, in hundreds of miles, between 30 international cities. Using the Python code below find the shortest airway system that connects all the cities, using only straight paths from one city to another.

```
import networkx as nx
import matplotlib.pyplot as plt
import numpy as np

dist_matrix = np.loadtxt('data/ha30_dist.txt')
positions = np.loadtxt('data/ha30_xyz.txt')
```

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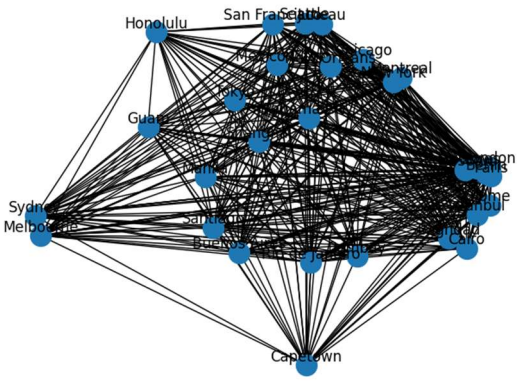
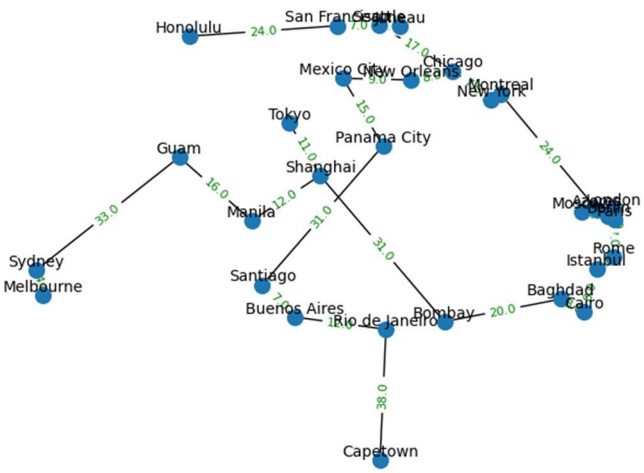
```
with open('data/ha30_name.txt') as f:
    city_list = f.readlines()

G_q3 = nx.Graph()
for i in range(len(city_list)):
    G_q3.add_node(city_list[i], pos=(positions[i, 0], positions[i,
1]))

for i in range(G_q3.number_of_nodes()):
    for j in range(i + 1, G_q3.number_of_nodes()):
        G_q3.add_edge(city_list[i], city_list[j],
weight=dist_matrix[i][j])

pos = nx.get_node_attributes(G_q3, 'pos')
nx.draw(G_q3, pos, with_labels=True)
plt.show()
```




```
T = nx.minimum_spanning_tree(G_q3)
pos = nx.get_node_attributes(T, 'pos')
labels = nx.get_edge_attributes(T, 'weight')
nx.draw(T, pos=pos, with_labels=True, font_size=10, node_size=100)
nx.draw_networkx_edge_labels(T, pos=pos, edge_labels=labels,
font_size=8, font_color='green')
plt.show()
```



```
edge_data = [
    {'source': src[:-1], 'destination': dest[:-1], 'distance':
dist['weight']}
    for src, dest, dist in T.edges(data=True)
]

display(pd.DataFrame(edge_data))

print("Total", sum([T[i][j]['weight'] for (i, j) in T.edges()]))
```

1-10 29 rows × 3 cols

	source	destination	distance
0	Azores	London	16.0
1	Azores	Montreal	24.0
2	Baghdad	Cairo	8.0
3	Baghdad	Bombay	20.0
4	Berlin	Paris	5.0
5	Berlin	Rome	7.0
6	Berlin	Moscow	10.0
7	Bombay	Shanghai	31.0
8	Buenos Aires	Santiago	7.0
9	Buenos Aires	Rio de Janeiro	12.0

Total 403.0

4. Find the non-isomorphic tree count of order 7 by using the Python code given below.

```
import networkx as nx
import matplotlib.pyplot as plt

order = 7
GList = [G for G in nx.nonisomorphic_trees(order)]
isom_count = len(GList)
print("Non-isomorphic tree count of order ", order, " is ",
isom_count)

fig, axes = plt.subplots(4, 3, figsize=(12, isom_count))
fig.delaxes(axes[-1, -1])
axes = axes.flat

for G, ax in zip(GList, axes):
    nx.draw(G, ax=ax, node_size=75)
plt.show()
```

Non-isomorphic tree count of order 7 is 11

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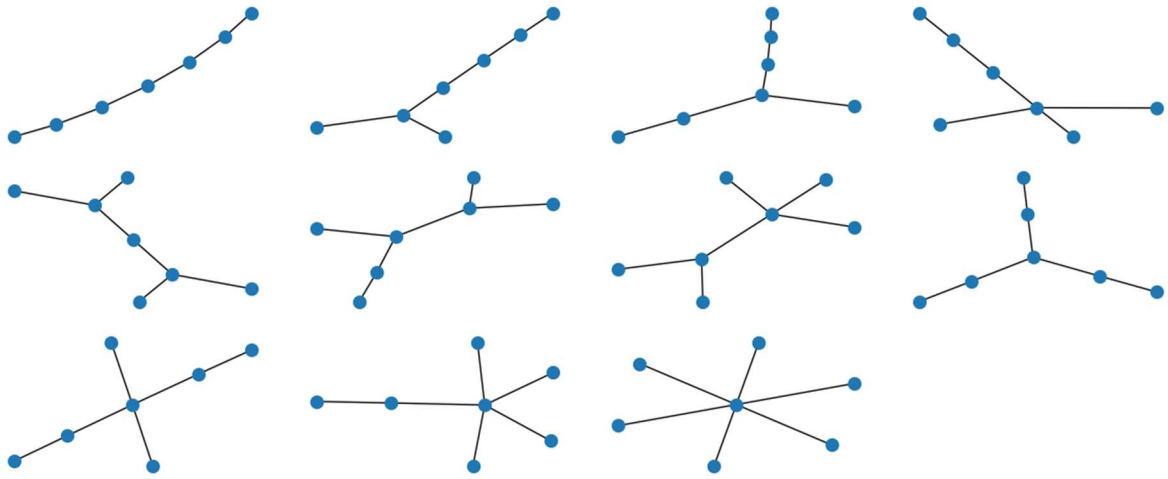
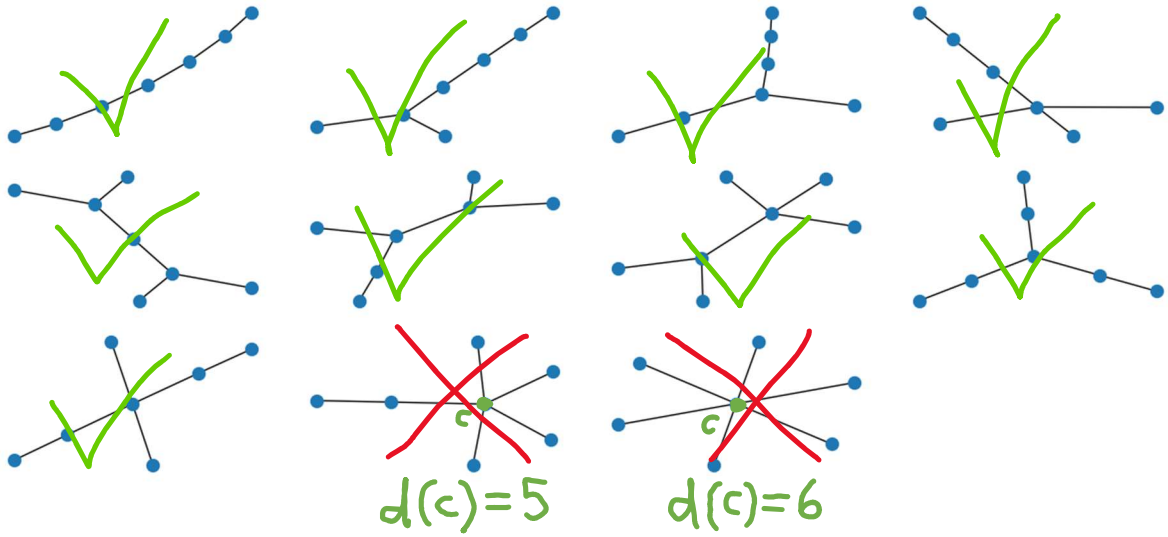


5. How many different isomers exist for the chemical formula C_nH_{2n+2} when n is 7.



There are 9 valid isomers for C_7H_{16}